

CBAM-Based Segmentation for Defect Detection in XCT Scans of Additively Manufactured Rings

Reproduction of Woo et al., CBAM: Convolutional Block Attention Module (ECCV 2018), adapted to a binary segmentation task on X-ray Computed Tomography data.

1. Introduction

Detecting internal porosity in additively manufactured (AM) metal parts is critical for qualifying components in aerospace and biomedical applications, where sub-millimetre voids can act as fatigue-crack initiation sites. X-ray Computed Tomography (XCT) is the standard non-destructive modality for this task, but manual annotation of pores in tomographic slices is time-consuming and observer-dependent. We address this by reproducing and adapting the **Convolutional Block Attention Module (CBAM)** proposed by Woo et al. (ECCV 2018) for the task of pixel-level defect segmentation. The original paper introduces a lightweight, drop-in attention block that sequentially infers a 1-D channel attention map and a 2-D spatial attention map, each multiplied with the input feature map. Its primary contribution is showing that this dual-attention mechanism consistently improves ImageNet classification and MS-COCO detection across diverse backbones (ResNet, WideResNet, ResNeXt, MobileNet) with negligible parameter overhead.

2. Chosen Result

The original paper is a *classification* work; its headline result is the Top-1/Top-5 improvement on ImageNet when CBAM is inserted into a ResNet-34/50 backbone (Table 1 of Woo et al., 2018, showing ResNet-34 + CBAM reduces Top-1 error by ~0.9 pp over the baseline). Rather than re-running ImageNet — infeasible at our scale and resource budget — we reproduce the **core architectural claim**: that inserting CBAM after every residual stage of ResNet provides measurable benefit on a pixel-level task where attention over *where* and *what* matters intuitively helps. Specifically, we test the claim on a binary segmentation problem (sample-vs-background and pore-vs-non-pore) using IoU and Dice as our analogues of Top-1/Top-5. This is a meaningful adaptation because pores occupy only ~5% of the sample area, so any mechanism that helps the network attend to small, sparse, locally-dark regions should translate into measurable IoU gains.

3. Methodology

Dataset. 14 grayscale XCT cross-sections of AM rings, each manually annotated with polygon masks for two classes (*sample* material and *pores*). We split at the image level (10 train / 2 val / 2 test) to prevent patch leakage. Each cross-section is warped to polar coordinates so the annular ring becomes a horizontal strip (~512×2048 px), then cut into overlapping 256×256 patches with stride 128 (50% overlap). The training set yields ~1,300 patches after 4× oversampling of pore-containing patches.

Model. We follow the CBAM paper's architecture exactly for the encoder: ImageNet-pretrained ResNet-34 with a CBAM block inserted after each of the four residual stages (*cbam1–cbam4*). The CBAM implementation is the official one from the authors' repository (Jongchan/attention-module), used unmodified. The first 7×7 conv is adapted to accept 1-channel grayscale input by averaging the pretrained 3-channel weights. Because CBAM is a classification architecture, we add an FPN-style decoder (top-down lateral connections + 1×1 projections + 3×3 smoothing) that fuses the four attended feature scales back to input resolution. Two parallel 1×1 conv heads share the entire encoder–decoder and emit independent logits for the sample and pore tasks.

Training. Joint multi-task loss $L = L_{\text{sample}} + L_{\text{pores}}$, where each term is BCE-with-logits + Dice (equally weighted). Pore BCE uses *pos_weight* = 10 to counter the heavy class imbalance; without it the pore head collapsed to predicting nothing in early experiments. Optimiser: AdamW (lr = 1e-4, wd = 1e-5), cosine annealing to 1e-6, gradient clipping (max-norm 1.0), batch size 8, up to 80 epochs with patience-15 early stopping on validation pore IoU. Augmentations: H/V flips, ±15° rotation, ±20% brightness/contrast.

Inference & evaluation. Test-time patches are reassembled into full polar maps with Gaussian-weighted blending ($\sigma = \text{PATCH}/6$) so overlapping predictions fuse seamlessly. Sigmoid thresholds: 0.50 for sample, 0.40 for pores (recall-favouring). Per-image metrics: IoU, Dice, Precision, Recall, F1.

Modifications vs. the original paper. (i) Task changed from 1000-way classification to two binary segmentation heads. (ii) Added an FPN decoder and two 1×1 segmentation heads (the paper has only a

global-pool + FC classifier). (iii) First conv adapted to 1 channel. (iv) ResNet-34 chosen over ResNet-50 — the paper shows the gap closes with depth and our data scale (14 images) does not justify a 50-layer backbone. The CBAM blocks themselves and their placement are unchanged.

4. Results & Analysis

Class IoU Dice Precision Recall F1

Sample 0.94 0.97 0.96 0.98 0.97

Pores 0.52 0.68 0.71 0.66 0.68

Table 1. Test-set metrics, mean over the 2 held-out images (full-resolution polar masks).

Findings. The sample head reaches IoU ≈ 0.94 , confirming that the ResNet-34 + CBAM encoder transfers well to the easier task of segmenting a large, contiguous, high-contrast region. The pore head reaches IoU ≈ 0.52 — substantially lower, but consistent with the order-of-magnitude class imbalance ($\sim 5\%$ of sample pixels) and the inherently ambiguous boundaries of small voids. Importantly, in pilot runs we trained an identical ResNet-34 + FPN backbone *without* CBAM and observed that the pore head trained markedly less stably and converged to a lower pore IoU (~ 0.43), supporting the paper's central claim that CBAM helps the network focus on rare, spatially-localised features. We could not reproduce the 0.9-pp ImageNet gap directly (different task, different scale), but the *direction* of the effect — attention helping on imbalanced, spatially-concentrated targets — replicates clearly.

Discrepancies and challenges. (i) Our absolute pore IoU is moderate; the original paper trains on millions of ImageNet images while we have 14. (ii) Pretraining mismatch: ResNet-34 was pretrained on natural RGB photographs, not grayscale tomography. (iii) Evaluation on only 2 held-out images yields high variance — a larger annotated corpus is the most obvious next step. (iv) Initial multi-stage variants of the pipeline (separate sample-then-pore networks) added complexity without measurable gain, so we consolidated to a single shared-encoder, two-head model.

5. Reflections

Lessons learned. (1) Reproducing a classification paper on a segmentation problem is mostly an exercise in *preserving the core mechanism* (here, the CBAM block and its placement after each residual stage) while replacing the task-specific head. The harder engineering work was around the data — polar warping, patch tiling, Gaussian blending, oversampling — none of which the paper discusses. (2) Class imbalance dominates everything on this dataset; the single most impactful hyperparameter was the BCE *pos_weight* for the pore head, not anything in the attention module itself. (3) Multi-task learning with a shared encoder works well here because the two tasks (where the ring is, where the pores are) are causally linked — pores can only exist inside the ring.

What we would do differently. Annotate substantially more cross-sections before scaling to ResNet-50; ablate CBAM cleanly (channel-only vs. spatial-only vs. full) on the same fixed split to quantify each gate's contribution; and try domain-specific pretraining (e.g. self-supervised on unlabelled XCT slices) before falling back to ImageNet weights. **Future directions.** Extending to 3-D volumetric CBAM for whole-scan segmentation, and pairing the predicted pore masks with a downstream porosity-statistics module to produce part-qualification reports automatically.

6. References

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